THE TWO DEFINITIONS OF H0 WITH COMPACT SUPPORT AGREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of H^0_c via functions with compact support (more-etale, Definition definition-compact-support, canonical tag 0F72). cited at statement line 4332. The first definition of the group; the lemma compares it with the second.
- Definition of H^0_c via cohomology of j_! (more-etale, Definition definition-cohomology-compact-support, canonical tag 0GJZ). cited at statement line 4333. The second definition, which by construction is H^0(Xbar, j_!F) for any compactification.
- Compactification formula for compact support (more-etale, Lemma lemma-compactify-compact-support, canonical tag 0F75; collected as card L22435). cited at proof lines 4340-4341. Shows the first version of H^0_c(X, F) equals H^0(Xbar, j_!F) for the chosen compactification.

Source reference. Original definition 0F72: sources/stacks-chow/tags.txt, source lines 18992–18992. Cited at statement line 4332: the first of the two definitions being compared.

Source reference. Original definition 0GJZ: sources/stacks-chow/tags.txt, source lines 20669–20669. Cited at statement line 4333: the second of the two definitions being compared.

Source reference. Original lemma 0F75: sources/stacks-chow/tags.txt, source lines 18995–18995. Cited at proof lines 4340-4341: identifies the first version with H0(Xbar, j_!F).

Lemma 0.2. *Let X be a separated scheme of finite type over a field k. If $\mathcal{F}$ is a torsion abelian sheaf, then the abelian group $H^0_c(X, \mathcal{F})$ defined in Definition 0F72 agrees with the abelian group $H^0_c(X, \mathcal{F})$ defined in Definition 0GJZ.*

Proof. Choose a compactification $j : X \to \overline{X}$ over k. In both cases the group is defined as $H^0(\overline{X}, j_!\mathcal{F})$. This is true for the first version by Lemma 0F75 and for the second version by construction. □

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